USER INFORMATION
READ CAREFULLY

INFUSOL® K 100mmol/L (STERILE POTASSIUM CHLORIDE CONCENTRATE INTRAVENOUS INFUSION 100mmol/L (0.746% w/v))
INFUSOL® K 200mmol/L (STERILE POTASSIUM CHLORIDE CONCENTRATE INTRAVENOUS INFUSION 200mmol/L (1.49% w/v))
INFUSOL® K 400mmol/L (STERILE POTASSIUM CHLORIDE CONCENTRATE INTRAVENOUS INFUSION 400mmol/L (2.98% w/v))

COMPOSITION:

Sterile Potassium Chloride Concentrate Intravenous Infusion (mmol Potassium /Container)	**Composition (g/L)** Potassium Chloride BP (KCl)	**Osmolarity* (mOsmol/L) (calc)**	**pH**	**Ionic Concentration (mmol/L)**	
				Potassium	Chloride
10 mmol/100 mL	7.46	**200**	5.0 (4.0 to 8.0)	**100**	100
10 mmol/50 mL	14.9	**400**	5.0 (4.0 to 8.0)	**200**	200
20 mmol/50 mL	29.8	**799**	5.0 (4.0 to 8.0)	**400**	400

*Normal physiologic osmolarity range is approximately 280 to 310 mOsmol/L. Administration of substantially hypertonic solutions (≥600 mOsmol/L) may cause vein damage.

PRODUCT DESCRIPTION:
Sterile Potassium Chloride Concentrate Intravenous Infusion is a clear and colourless solution, sterile, non-pyrogenic, highly concentrated, ready-to-use, solution of Potassium Chloride BP in Water for Injections for electrolyte replenishment in a single dose container for intravenous administration. It contains no antimicrobial agents. Ready-to-use solution for central route. Must be further diluted if peripheral line is used with INFUSOL® K 200mmol/L and INFUSOL® K 400mmol/L.

PHARMACODYNAMICS:
Potassium is the major cation of body cells (160 mmol/liter of intracellular water) and is concerned with the maintenance of body fluid composition and electrolyte balance. Potassium participates in carbohydrate utilization, protein synthesis, and is critical in the regulation of nerve conduction and muscle contraction, particularly in the heart. Chloride, the major extracellular anion, closely follows the metabolism of sodium, and changes in the acid-base of the body are reflected by changes in the chloride concentration.

PHARMACOKINETICS:
Normally about 80 to 90% of the potassium intake is excreted in the urine, the remainder in the stools and to a small extent, in the perspiration. The kidney does not conserve potassium well so that during fasting, or in patients on a potassium-free diet, potassium loss from the body continues resulting in potassium depletion. A deficiency of either potassium or chloride will lead to a deficit of the other.

INDICATION:
Sterile Potassium Chloride Concentrate Intravenous Infusion is indicated in the treatment of potassium deficiency states when oral replacement is not feasible.
When using these products, these patients should be on continuous cardiac monitoring and frequent testing for serum potassium concentration and acid-base balance.

RECOMMENDED DOSAGE:
The dose and rate of administration are dependent upon the specific condition of each patient.

INFUSOL® K 200mmol/L and INFUSOL® K 400mmol/L MUST BE FURTHER DILUTED IF PERIPHERAL LINE IS USED. For example, 50mL of INFUSOL® K 200mmol/L or 25 mL of INFUSOL® K 400mmol/L dilute with 100mL of Sodium Chloride 0.9% w/v Intravenous Infusion BP or 100mL of Dextrose (Glucose) 5.0% Infusion BP and mixed well.

Administer intravenously only with a calibrated infusion device at a slow, controlled rate. Because pain associated with peripheral infusion of Potassium Chloride solution has been reported, whenever possible administration via a central route is recommended for thorough dilution by the blood stream and avoidance of extravasation. Highest concentrations (300 and 400 mmol/L) should be exclusively administered via central route.

Recommended administration rates should not usually exceed 10 mmol/hour or 200 mmol for a 24 hours period if the serum potassium level is greater than 2.5 mmol/liter.

In urgent cases where the serum potassium level is less than 2 mmol/liter or where severe hypokalemia is a threat (serum potassium level less than 2 mmol/liter and electrocardiographic changes and/or muscle paralysis), rates up to 40 mmol/hour or 400 mmol over a 24 hours period can be administered very carefully when guided by continuous monitoring of the EKG and frequent serum K^+ determinations to avoid hyperkalemia and cardiac arrest.

Parenteral drug products should be inspected visually for particulate matter and discolouration, whenever solution and container permit. Use of a final filter is recommended during administration of all parenteral solutions where possible. **Do not add supplementary medication.**

ROUTE OF ADMINISTRATION: Intravenous (IV)

CONTRAINDICATIONS:
Sterile Potassium Chloride Concentrate Intravenous Infusion is contraindicated in diseases where high potassium levels may be encountered, and in patients with hyperkalemia, renal failure and in conditions in which potassium retention is present.

WARNINGS:
THIS HIGHLY CONCENTRATED, READY-TO-USE STERILE POTASSIUM IS INTENDED FOR THE MAINTENANCE OF SERUM K^+ LEVELS AND FOR POTASSIUM SUPPLEMENTATION IN FLUID RESTRICTED PATIENTS WHO CANNOT ACCOMMODATE ADDITIONAL VOLUMES OF FLUID ASSOCIATED WITH POTASSIUM SOLUTIONS OF LOWER CONCENTRATION.
TO AVOID POTASSIUM INTOXICATION, DO NOT INFUSE THESE SOLUTIONS RAPIDLY. PATIENTS REQUIRING HIGHLY CONCENTRATED SOLUTIONS SHOULD BE KEPT ON CONTINUOUS CARDIAC MONITORING AND UNDERGO FREQUENT TESTING FOR SERUM POTASSIUM AND ACID-BASE BALANCE, ESPECIALLY IF THEY RECEIVE DIGITALIS.
In patients with renal insufficiency, administration of potassium chloride may cause potassium intoxication and life threatening hyperkalemia.
Administer intravenously only with a calibrated infusion device at a slow, controlled rate (See RECOMMENDED DOSAGE). Because pain associated with peripheral infusion of Potassium Chloride solution has been reported, whenever possible administration via a central route is recommended for thorough dilution by the blood stream and avoidance of extravasation. Highest concentrations (300 and 400 mmol/L) should be exclusively administered via central route.
The administration of intravenous solutions can cause fluid and/or solute overload resulting in dilution of serum electrolyte concentrations, overhydration, congested states or pulmonary edema. The risk of dilutional states is inversely proportional to the electrolyte concentration. The risk of solute overload causing congested states with peripheral and pulmonary edema is directly proportional to the electrolyte concentration.

PRECAUTIONS:
Single dose container. Discard all unused contents. Do not use if leakage is detected. If any visible solid particles, growth or turbidity appears during storage, the product should not be used.

Laboratory Tests:
Serum potassium levels are not necessarily indicative of tissue potassium levels. Solutions containing potassium should be used with caution in the presence of cardiac or renal disease. Clinical evaluation and periodic laboratory determinations are necessary to monitor changes in fluid balance, electrolyte concentrations, and acid-base balance during prolonged parenteral therapy or whenever the condition of the patient warrants such evaluation. Significant deviations from normal concentrations may require the use of additional electrolyte supplements, or the use of electrolyte-free dextrose solutions to which individualized electrolyte supplements may be added.

Pregnancy:
Pregnancy Category C. Animal reproduction studies have not been conducted with potassium chloride. It is also not known whether potassium chloride can cause fetal harm when administered to a pregnant woman or can affect reproduction capacity. Potassium chloride should be given to a pregnant woman only if clearly needed.

Pediatric Use:
These products should not be used in pediatric patients at this time.

INTERACTION WITH OTHER MEDICAMENTS:
Sterile Potassium Chloride Concentrate Intravenous Infusion should be used with caution in patients treated concurrently or recently with agents or products that can cause hyperkalemia or increase the risk of hyperkalemia (e.g.: ACE inhibitors and potassium-sparing diuretics (e.g. Amiloride, Spironolactone, Triamterene)). Administration of potassium in patients treated with such agents is associated with an increased risk of severe and potentially fatal hyperkalemia, in particular in the presence of other risk factors for hyperkalemia.

Sterile Potassium Chloride Concentrate Intravenous Infusion is not recommended with concurrent use of digitalis glycosides in those patients with severe or complete heart block. Careful monitoring is necessary if potassium chloride is used to correct hypokalaemia in such patients.

ADVERSE EFFECTS:
Potassium intoxication with mild or severe hyperkalemia has been reported. The signs and symptoms of intoxication include paresthesia of the extremities, areflexia, muscular or respiratory paralysis, mental confusion, weakness, hypotension, cardiac arrhythmia, heart block, electrographic abnormalities and cardiac arrest. EKG abnormalities serve as a clinical reflection of the seriousness of changes in serum potassium concentrations: peaked T waves and prolonged P-R intervals usually occur with modest elevations above the upper limit of normal potassium concentrations; P waves disappear, the QRS complex widens, and eventual asystole usually occurs with higher elevations.
Reactions which may occur because of the solution or the technique of administration include febrile response, infection at the site of injection, venous thrombosis or phlebitis extending from the site of injection, extravasation and hypervolemia. Infusion of highly concentrated potassium chloride solutions may cause local pain and vein irritation (See WARNINGS).
Reactions reported with the use of potassium-containing solutions include nausea, vomiting, and abdominal pain and diarrhea.
If an adverse reaction does occur, discontinue the infusion, evaluate the patient, institute appropriate therapeutic counter measures and save the remainder of the fluid for examination if deemed necessary.

OVERDOSE AND TREATMENT:
In the event of hyperkalemia, discontinue the infusion immediately and institute corrective therapy to reduce serum potassium levels as necessary. The use of potassium containing foods or medications must also be eliminated. Treatment of mild to severe hyperkalemia with signs and symptoms of potassium intoxication includes the following:
1. Dextrose Injection, 10% or 25%, containing 10 units of crystalline insulin per 20 grams of dextrose administered intravenously, 300 to 500mL per hour.
2. Absorption and exchange of potassium using sodium or ammonium cycle cation exchange resin, orally and as retention enema.
3. Hemodialysis and peritoneal dialysis.
In cases of digitalization, too rapid a lowering of plasma potassium concentration can cause digitalis toxicity.

INCOMPATIBILITIES:
The compatibility of any additives should be checked before use.

STORAGE CONDITION BEFORE AND AFTER DILUTION: Do not store above 30°C.

SHELF LIFE:
3 years from manufacturing date in the proposed storage condition. The solution should be used within 24 hours after dilution.
Do not use after expiry.

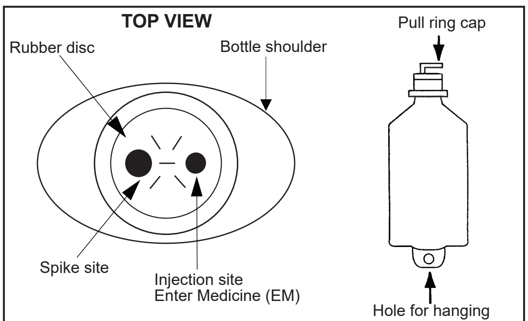
DOSAGE FORM AND PACKAGING AVAILABLE:
INFUSOL® K 100mmol/L: 100mL x 20 LDPE plastic bottle per carton
INFUSOL® K 200mmol/L: 50mL x 20 LDPE plastic bottle per carton
INFUSOL® K 400mmol/L: 50mL x 20 LDPE plastic bottle per carton

MANUFACTURER / PRODUCT REGISTRATION HOLDER:

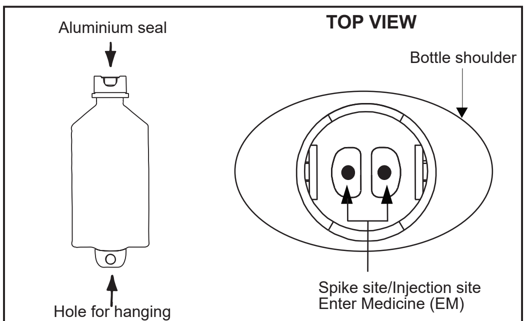
AIN MEDICARE SDN.BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

HANDLING INSTRUCTION

AIN MEDICARE INTRAVENOUS INFUSION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) BOTTLE
The bottle is made from injectable grade polyethylene.
PARTS OF THE PLASTIC BOTTLE

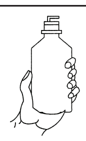


Design 1: Pull Ring Cap



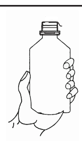
Design 2: Twin Port Cap (Aluminum Seal)

Top view after pull ring cap and aluminium seal is removed. Please refer whichever applicable design.
INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG

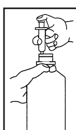


Design 1

1. Before use check for any solid particle, growth, turbidity or leakage. Discard the product if particle, growth, turbidity or leakage is found.

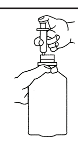


Design 2

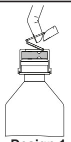


Design 1

3. Additives may be injected through injection site (Refer to Design 1 or 2 whichever applicable) on the rubber. Mix the solution thoroughly.



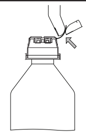
Design 2



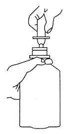
Design 1

2. Design 1: Pull the cap's ring until completely removed.

Design 2: Pull the aluminum seal until completely removed by centre line.



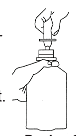
Design 2



Design 1

4. Hold bottle down firmly and insert a spike of administration set into the spike site for Design 1 or for Design 2, pull the other aluminium seal until completely removed before inserting a spike of administration set.

(Spike site is meant for spiking of the administration set and is only for single use)



Design 2

Date of Revision: 01.07.2022